

CALL FOR PAPERS

IEEE Transactions on VLSI Systems Special Section on Low Power Electronics and Design

Power consumption is a critical constraint influencing the design and operation of a wide variety of computing and communication systems. Power-related metrics have become important for different reasons, such as reducing cooling and packaging costs, prolonging battery life, and increasing reliability of the system. Power optimization requires synergistic approaches at different levels of abstraction, ranging from identifying new technologies and circuit styles, to designing power-aware architectures, systems, and software. This special section will report on recent advances in development of power-efficient and power-aware circuits and systems. Topics of interest include, but are not limited to:

- *Technologies and Digital Circuits.* Emerging logic and memory technologies, Device design, Low power low leakage circuits, Memory circuits, Cooling technologies, Battery technologies.
- *Logic and Microarchitecture Design.* Processor core design, Cache design, Logic and RTL design, Arithmetic and signal processing circuits, Encryption technologies, Asynchronous design.
- *Analog, MEMS and Mixed Signal Electronics.* RF circuits, Wireless, MEMS circuits, AD/DA Converters, Mixed-signal circuits, DC-DC conversion.
- *Design Tools.* Energy simulation and estimation tools that operate at the circuit/gate level, RT level, behavioral level, and algorithmic level, Physical design and interconnects.
- *System Design and Methodologies.* Microprocessor, DSP and embedded systems design, FPGA and ASIC designs, System level power management, Behavioral and system level design aids.
- *Software Design and Optimization.* Power aware compiler and operating system design, Application level optimizations, Wireless and sensor networks.
- *Power-aware Applications* like sensor networks, personal mobile computation/communication, ...

Authors are encouraged to submit high-quality research contributions that will not require major revisions. Extensions of papers presented at the ISLPED'06 – the International Symposium on Low Power Electronics and Design, Tegernsee, Germany, Oct.4-6, 2006 (<http://www.islped.org>) are especially encouraged. Please identify clearly the additional material from the original symposium paper in your submitted manuscript. Submissions of relevant original work not presented at ISLPED'06 are also welcome. All manuscripts are subject to standard IEEE Transactions on VLSI review process. Prospective authors should submit their manuscripts electronically on the TVLSI Web site: <http://tvlsi-ieee.manuscriptcentral.com/>

Authors should clearly identify their papers as submissions for the “Special Section on Low Power Electronics and Design 2007” on their manuscript and in the Note to Editor field of the web submission form. Instructions on how to submit a paper can be found at <http://tvlsi-ieee.manuscriptcentral.com/> and authors can contact Michael Pham at mpham@csee.usf.edu for further assistance.

Important Dates:

Submission Deadline:	February 2, 2007
Notification Date:	May 4, 2007
Final Version Due:	June 1, 2007

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